



Numerical Cognition in Mandarin-Speaking & English-Speaking 2-to-4-year-old Children in China and the U.S.

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Introduction

Background

- ❖ Children begin to count at around age two, but representing exact set sizes (subset-knowers) and generating any set size (cardinal principle knowers) is more protracted.
- ❖ However, numerical cognition can also be influenced by language and culture.
- ❖ Few studies have looked into the development from subset-knowers to CP-knowers in Mandarin speaking children.
 - According to the morphological bootstrapping hypothesis, Chinese does not have singular/plural distinctions in nouns, leading to **delayed acquisition** of the meaning of the number word for one⁶.
 - Different and maturational influences affect different aspects of children's developing mathematical competencies⁶.
- ❖ Knower-level theory: children progress through a series of stages in numerical development, each characterized by a different level of understanding and presenting numbers (e.g. a one-knower only understands the concept of “one”). Before a child becomes a cardinal principle knower (CP knower), they could only learn the meaning of number words within the range of parallel individuation one by one, in sequential order.

Research Questions

- ❖ What role does language play in numerical cognition development in young children?
 - Do children growing up speaking Mandarin Chinese become one-knowers later than English-speaking children?
- ❖ How will English- and Mandarin- speaking children perform similarly or differently in the numerical cognition tasks? is numerical cognition development universal cross language and cultures?

Methods

Participants

- ❖ 84 Mandarin-speaking children (37 F & 47 M, mean age = 42 months, range = 24-59 months)
- ❖ 53 English-speaking children (24 F & 24 M, mean age = 40.3 months, range = 25-56 months) from a previous study conducted in our lab.

Procedure

- ❖ Tasks were adapted from previous English study, Pirahã batteries^{3, 4}.
 - Ascending & Descending Enumeration Task
 - Give-N Task⁷
 - Parallel & Orthogonal One-to-One Correspondence Task
 - Nuts-in-a-Can Task
 - Rote Counting Task
- ❖ Caregiver Interview

Results

Age & Knower Level Distribution

- ❖ Number Knower Level was determined by the Given-N Task. The child was denoted as a knower of the highest number with which they had 66% accuracy.

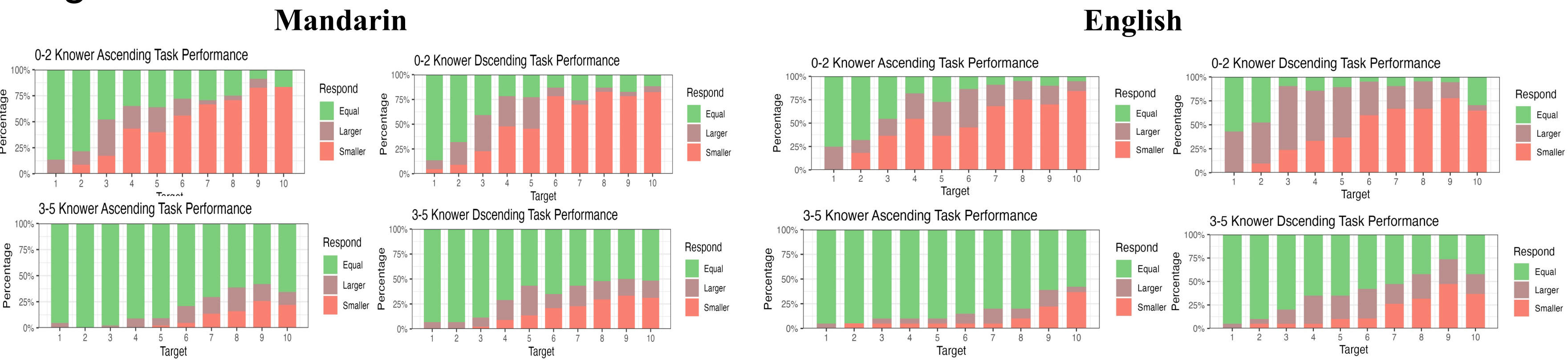
Mandarin				
Age Group (Y)	N	%	Age (Median, IQR)	Rote Counting (Median, IQR)
2~3	19	24.1%	39, 31~47	10.5, 9~12
3~4	39	49.4%	37, 29~47	10, 8~10
4~5	21	26.6%	37, 27~48	10, 1~48
English				
Age Group (Y)	N	%	Age (Median, IQR)	Rote Counting (Median, IQR)
2~3	10	20.8%	39, 34~48	5, 3~69
3~4	27	56.3%	38, 28~53	20, 10~39
4~5	11	22.9%	49, 36~59	24.5, 4~190

Mandarin				
Knower Level	N	%	Age (Median, IQR)	Rote Counting (Median, IQR)
Non-Knower	2	2.7%	36, 30~36	8, 6~11
One-Knower	5	6.7%	38, 36~40	12, 11~14
Two-Knower	19	25.3%	38, 36~47	9, 6~10
Three-Knower	9	12.0%	44, 43~45	10, 10~10
Four-Knower	10	13.3%	36, 36~45	10, 8~11
Five-Knower	30	40.0%	44, 40~48	20, 11~30

English				
Knower Level	N	%	Age (Median, IQR)	Rote Counting (Median, IQR)
Non-Knower	7	15.2%	36, 30~36	8, 6~11
One-Knower	6	13.0%	38, 36~40	12, 11~14
Two-Knower	11	23.9%	38, 36~47	9, 6~10
Three-Knower	2	4.3%	44, 43~45	10, 10~10
Four-Knower	5	10.9%	36, 36~45	10, 8~11
Five-Knower	15	32.6%	44, 40~48	20, 11~30

Ascending & Descending Enumeration Task

- ❖ Verbal. Blocks were laid out one at a time (1 to 10) and the child were asked to say how many there were.
- ❖ Children's performance in this task was divided into none- to two-knower (subset knower) and three- to five-knower (CP-knower), as shown in the figure below.



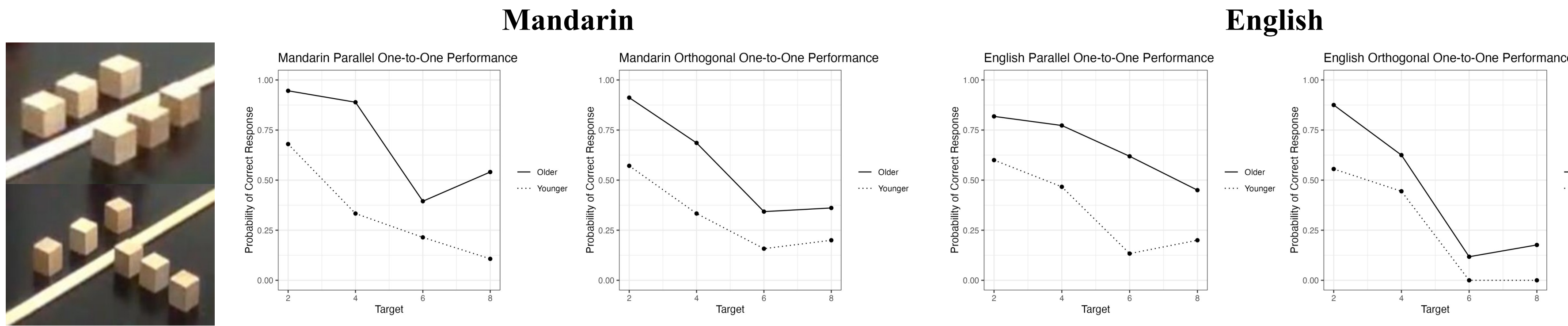
- ❖ Younger children and subset knower performed worse in both Ascending and Descending Tasks in both Mandarin and English groups.
- ❖ Both English- and Mandarin- speaking children performed **significantly worse** in the Descending Task compared to the Ascending task.

Mandarin					English						
Task	Group	Mean Pc	p-value (Group)	Mean Pc (Task)	p-value (Task)	Task	Group	Mean Pc (Age)	p-value (Group)	Mean Pc (Task)	p-value (Task)
Ascending	Younger	0.48	< 0.0001***	0.66	0.033*	Ascending	Younger	0.34	< 0.0001***	0.56	0.013*
	Older	0.81					Older	0.74			
	Subset Knower	0.38	< 0.0001***				Subset Knower	0.30	< 0.0001***		
	CP Knower	0.82					CP Knower	0.84			
Descending	Younger	0.40	0.0002***	0.56	0.013*	Descending	Younger	0.24	0.002**	0.39	< 0.0001***
	Older	0.69					Older	0.51			
	Subset Knower	0.34	< 0.0001***				Subset Knower	0.18	< 0.0001***		
	CP Knower	0.68					CP Knower	0.61			

Parallel & Orthogonal One-to-One Correspondence Task

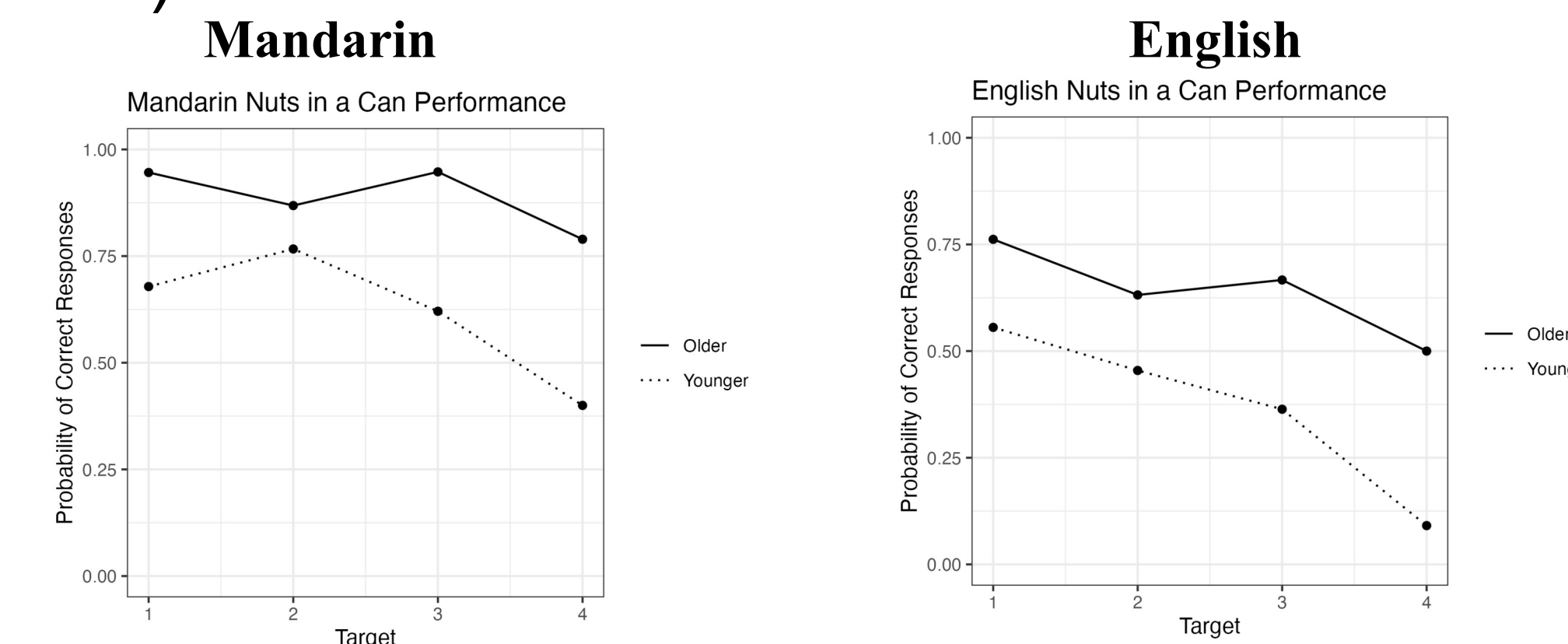
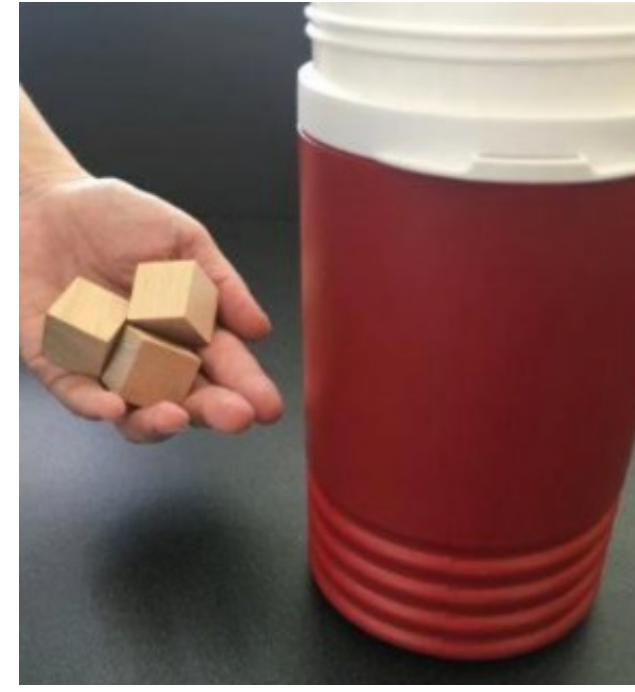
- ❖ Non-verbal. Children were asked to lay out the same number of blocks (n=2,4,6,8) on their side of divider as that modeled by the experimenter.
- ❖ English- and Mandarin-speaking children **did not show significant differences** in performance of these tasks. (t = -0.5, p = 0.62)
 - However, English and Mandarin with group comparison shows that English speaking children performed **significantly worse** in Orthogonal Task (t = -2.23, p = 0.015), while Mandarin speaking children do not show this pattern (t = -1.02, p = 0.15).

- ❖ Both younger and older children performed worse when the target got larger in both Parallel and Orthogonal Tasks (see figures).



Nuts-in-a-Can Task

- ❖ Verbal. The experimenter showed children a set of blocks (n = 1,2,3,4) and put them in a container. Blocks were removed one at a time and the child was asked if there were any more left inside the container.
- ❖ Mandarin group performed **significantly better** than the English group in this task. (t = -3.03, p = 0.003)
- ❖ Both English and Mandarin showed a downward progressive trend as the target gets larger. (see figures)



Discussion & Conclusions

- ❖ Mandarin- and English- speaking children performed **similarly** for majority of verbal and non-verbal tasks. Some aspects of numerical cognition development are likely universal across languages and cultures.
- ❖ Our results **contradicted** previous research that reported delays in numerical cognition acquisition in Japanese- and Mandarin children.
- ❖ Age was a stronger predictor of children's performance compared to the number-knower level.
- ❖ Children's ability to rote counting was not a predictor of their performance. Relying solely on rote counting may not accurately assess children's ability.

References

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